LEGS-Bench: Toward a Unified Evaluation Protocol for Language-Embedded 3D Gaussian Splatting in Embodied Agents

Anonymous Submission

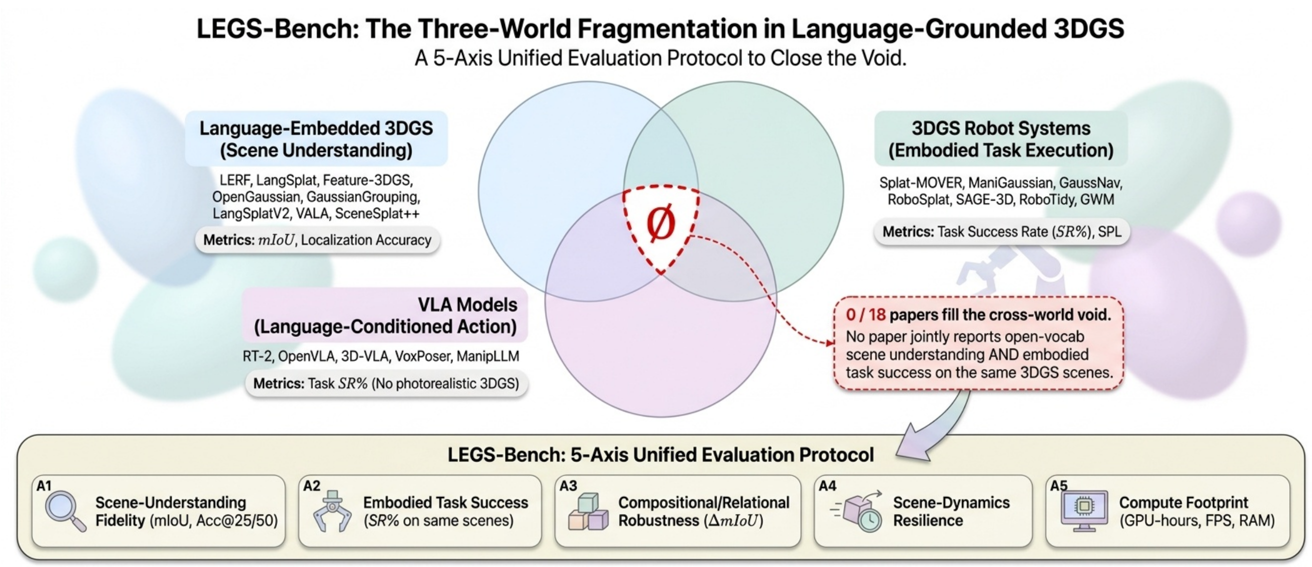

Figure 1. The three disjoint literatures that LEGS-Bench aims to unify. *Left*: language-embedded 3DGS methods that report scene-understanding quality only. *Center*: 3DGS-based robot systems that report task success with categorical language. *Right*: VLA models with standardized task benchmarks but no 3DGS representation. No paper today fills all five axes of our proposed protocol on the same scene instances.

Abstract

Language-embedded 3D Gaussian Splatting (3DGS) sits at the intersection of vision-language models and robot policies, yet the field has fragmented into three disjoint literatures: scene-understanding methods that distill CLIP/SAM features into Gaussians [14, 21, 28, 38, 43, 49], 3DGS robot systems with custom task protocols [6, 19, 24, 30, 32, 39], and VLA models with standardized benchmarks but no 3DGS representation [3, 16, 47]. We audit twenty-two 2024–2026 papers against a five-axis evaluation matrix and find that no row is complete: not a single paper jointly reports open-vocabulary scene-understanding quality and *downstream embodied-task success on the same scene instances. The result is robust to dropping the two least-adopted axes. We propose LEGS-Bench, a unified evaluation protocol whose five axes—open-vocabulary localization accuracy, downstream task success, compositional query robustness, scene-dynamics resilience, and compute footprint—are designed to be reported together on a shared scene set. Our contribution is the gap, the protocol, and a meta-analysis that operationalizes both.*

1. Introduction

A robot told "bring me the red mug to the left of the kettle" must build a persistent 3D scene representation, ground an open-vocabulary phrase to a region, and condition a policy on the result. Language-embedded 3DGS [13]—which distills CLIP/SAM/DINO into per-Gaussian descriptors [14, 28, 43, 49]—is the natural bridge. In practice, three communities have grown around 3DGS in parallel and do not evaluate on the same things.

World 1: scene-understanding. LERF [14], LangSplat [28], Feature 3DGS [49], OpenGaussian [43], Gaussian Grouping [46], GARField [15], N2F2 [2], Semantic Gaussians [9], LangSplatV2 [21], VALA [42], SceneSplat++ [38], OpenSplat3D [27], SLAG [41], and FineSplat [36] all report open-vocabulary localization or mIoU on LERF-OVS/3D-OVS. *None reports a robot task success rate.*

World 2: 3DGS robot systems. Splat-MOVER [32], ManiGaussian [24]/++ [35], Splat-Nav [6], GaussNav [19], LERF-TOGO [30], GWM [25], EmbodiedSplat [7], and RoboSplat [45] report task success on custom suites with categorical labels. *None reports a scene-understanding metric on the same scenes.*

World 3: VLA models. RT-2 [3], OpenVLA [16], 3D-VLA [47], and VoxPoser [10] have disciplined task benchmarks but use 2D pixels, point clouds, or voxels—not 3DGS.

The empirical void. No paper jointly reports open-vocabulary scene-understanding quality and downstream embodied-task success on the same scene instances. We establish this across twenty-two 2024–2026 papers. Even SceneSplat-Bench [37], SAGE-Bench [40], and RoboTidy [39] each cover at most two of five axes.

LEGS-Bench. We propose a five-axis reporting protocol: (1) open-vocabulary scene-understanding quality, (2) downstream task success, (3) compositional and relational query robustness, (4) scene-dynamics resilience, and (5) compute footprint—all on the same scene instances.

Contributions. (i) A meta-analysis across twenty-two papers (§4) establishing the cross-thread evaluation void. (ii) The five-axis LEGS-Bench protocol (§3) reportable on a shared scene set. (iii) A discussion of annotation and infrastructure still needed—compositional-query annotations and a dynamics harness for Habitat-GS [44]. We train no new models; the question "which language-3DGS method is best for embodied agents?" is today ill-posed, and LEGS-Bench is the minimal protocol for making it well-posed.

2. Related Work

Language-embedded 3DGS (World 1). LERF [14] distills CLIP/DINO into a NeRF; the field shifted to 3DGS [13]: LangSplat [28] (CLIP autoencoder), Feature 3DGS [49] (general features), OpenGaussian [43] (point-level queries), Gaussian Grouping [46] (SAM codes), GARField [15], N2F2 [2], Semantic Gaussians [9], Lift3D [34], ReasonGrounder [23]. The 2025–2026 frontier: LangSplatV2 [21] (450+ FPS), VALA [42], SceneSplat++ [38] (1060 scenes), OpenSplat3D [27], Identity-aware LGS [12], SLAG [41] (city-scale), FineSplat [36] (part-level). *None reports a robot success rate.*

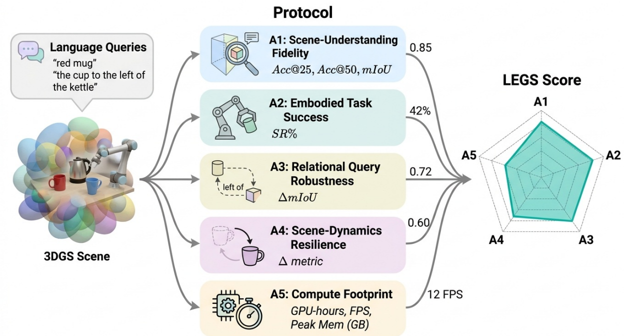

Figure 2. LEGS-Bench five-axis protocol. All axes are evaluated on the same shared scene pool $\mathcal{P}$.

3DGS in robot systems (World 2). Manipulation: Splat-MOVER [32], ManiGaussian [24]/++ [35] (world model), GWM [25] (future rollout), RoboSplat [45] (demonstration synthesis). Navigation: Splat-Nav [6], GaussNav [19], 3DGSNav [48], EmbodiedSplat [7]. SAGE-3D [40] (2M VLN tasks), RoboTidy [39] (500 scenes), GSMem [26], GaussExplorer [18]. *Language enters as categorical labels; no scene-understanding metric on the same scenes.*

VLA models (World 3). RT-1 [4], RT-2 [3], OpenVLA [16], 3D-VLA [47], VoxPoser [10], and ManipLLM [20] provide disciplined task evaluations (RLBench, Open-X, VLN-CE [17]) but use 2D pixels, point clouds, or voxels—no 3DGS.

Benchmark gap. Scene-understanding benchmarks (LERF-OVS, 3D-OVS [22], Replica [33], ScanNet [8]/200 [31], SceneSplat-Bench [37]) omit action metrics. Embodied benchmarks (HM3D [29], MP3D [5], RLBench [11], SAGE-Bench, RoboTidy) omit scene-understanding metrics on the same scenes. SAGE-Bench and RoboTidy come closest but use fixed vocabularies. LEGS-Bench proposes the standard that makes these columns comparable.

3. The LEGS-Bench Protocol

LEGS-Bench is an evaluation protocol, not a model. It specifies five reporting axes characterizing whether a language-embedded 3DGS representation is useful for embodied agents. All axes are measurable on existing methods without retraining—only re-evaluation on a shared scene pool. This paper ships the protocol *specification*; a public scene-pool release, scoring scripts, and evaluation harness are follow-up work (§3.6).

Notation. A scene $\mathcal{S} = \{g_i\}_{i=1}^{N}$ is a set of optimized Gaussians with language features $\mathbf{f}_i \in \mathbb{R}^d$. A query q is a natural-language string; $\Phi(q, \mathcal{S})$ returns a 3D mask, 2D mask, or target pose. A scene pool $\mathcal{P} = \{\mathcal{S}_1, \dots, \mathcal{S}_K\}$ is shared across all five axes.

Scoring rubric for the meta-analysis. For Tab. 1 we score each (paper, axis) cell as: ✓ = a numerical result reported on the paper's claimed scenes for this axis, with the protocol that axis specifies (e.g., open-vocabulary mIoU for A1, success rate for A2); △ = partial—the paper reports the axis but in a relaxed form (closed-vocabulary where open-vocabulary is required; training time without query latency for A5; or the axis is reported but on different scenes than its A1 row); ✗ = unreported. Reclassifying a △ to 3 requires (i) matching protocol and (ii) overlapping scene set with the same paper's other axes.

3.1. A1: Scene-Understanding Fidelity (SUF)

For each scene $\mathcal{S}_k$: $\text{Acc}@\tau = \frac{1}{|\mathcal{Q}_k|}\sum_q \mathbb{1}[\text{IoU}(\Phi(q, \mathcal{S}_k), M_q^\star) \geq \tau]$ for $\tau \in \{0.25, 0.50\}$, and $\text{mIoU}_k = \frac{1}{|\mathcal{Q}_k|}\sum_q \text{IoU}(\Phi(q, \mathcal{S}_k), M_q^\star)$. Report $(\text{Acc}@25, \text{Acc}@50, \text{mIoU})$ averaged over $\mathcal{P}$.

3.2. A2: Embodied Task Success (ETS)

ETS must be measured on the **same** scenes $\mathcal{P}$ as A1 (the structural gap in the literature). A submission reports at least one of: (1) open-vocabulary grasp ($\geq$20 queries/scene, success = object lifted $\geq$5 cm); (2) text-goal navigation (SR_{nav}, SPL [1]); or (3) text-goal retrieval (SR_{ret}). A1 without any A2 is non-compliant.

3.3. A3: Compositional and Relational Query Robustness (CRQR)

Four categories: **Base** (`[noun]`); **Attribute** (`[adj noun]`); **Spatial** (`[noun rel noun]`); **Negation/part**. Robustness: $\Delta_c = (\text{mIoU}_{\text{base}} - \text{mIoU}_c)/\text{mIoU}_{\text{base}}$, reported for all four. Categories target documented failure modes: attribute-binding and spatial-reference errors recur in OpenVLA [16] and RT-2 [3] traces; negation/part queries break CLIP-pooled features in VoxPoser [10]-class systems.

3.4. A4: Dynamics Resilience (DR)

DR is a *simulation-based* stress test: (i) train the language-3DGS field on the static scene; (ii) record baseline A1+A2 metrics; (iii) script-displace $k \in \{1, 2, 5\}$ rigid objects by $\geq$30 cm and re-render from the same trajectory; (iv) re-evaluate A1+A2 *without* re-optimizing the field. Report $\text{DR}^{\text{A1}}(k) = (\text{mIoU}^{(k)} - \text{mIoU}^{(0)})/\text{mIoU}^{(0)}$ and the analogous $\text{DR}^{\text{A2}}(k)$. Any update strategy (e.g., per-object re-anchoring) must be declared.

3.5. A5: Compute Footprint (CF)

Four numbers per scene: training cost T_{train} (H100-equiv. GPU-h), query latency ℓ_{query} (ms, batch 1, fp16), peak GPU memory Mem_{peak} (GB), stored size (GB). H100-equivalent hours require disclosing the TFLOPS conversion factor.

3.6. Score Vector and Recommended Scene Pool

The score tuple $\text{LEGS}(\mathcal{M}, \mathcal{P}) = (\text{SUF}, \text{ETS}, \text{CRQR}, \text{DR}, \text{CF})$ is not collapsed into a scalar—heterogeneous trade-offs matter for deployment. We recommend ten scenes: five LERF-OVS [14] (masks available) and five Habitat-GS [44] rooms. The release plan covers a public scene-ID manifest, CRQR query corpus, scoring scripts, and an evaluation harness; these are explicitly future work.

4. Meta-Analysis of the Coverage Gap

We score twenty-two 2024–2026 papers against the LEGS-Bench protocol (§3): eleven scene-understanding methods (LERF [14], LangSplat [28], Feature 3DGS [49], Open-Gaussian [43], Gaussian Grouping [46], SceneSplat++ [38], LangSplatV2 [21], VALA [42], OpenSplat3D [27], FineSplat [36], SLAG [41]), eight 3DGS-robotics methods (Splat-MOVER [32], LERF-TOGO [30], ManiGaussian++ [35], GWM [25], GaussNav [19], 3DGSNav [48], SAGE-3D [40], RoboTidy [39]), and three VLA models (RT-2 [3], Open-VLA [16], 3D-VLA [47]).

4.1. Coverage Matrix

Table 1 cross-tabulates the twenty-two papers against the five LEGS axes. ✓ = reported with concrete numbers on the paper's claimed scenes; △ = partial; ✗ = unreported (full rubric: §3).

Quantitative summary. The 110-cell matrix (22×5) has 27 fully-compliant (✓) cells (≈24.5%). Per-axis: A1 SUF 12/22; A2 ETS 11/22; A3 CRQR 1/22 (FineSplat only); A4 DR **0/22**; A5 CF 3/22. Rows complete on all five axes: **0 of 22**. Co-located A1+A2 on the same scenes: still zero. LERF-TOGO and Splat-MOVER come closest but report neither A3, A4, nor A5.

Goalpost robustness. A3 and A4 are not yet community-adopted, so 0/22 on those axes partly reflects measurement-novelty. On the *robust subset* A1+A2+A5, no paper still completes all three on one shared scene set: LERF-TOGO and Splat-MOVER omit A5; LangSplatV2, SceneSplat++, and SLAG report A1+A5 but not A2. The structural gap survives the most generous rubric restriction.

Paper	A1 SUF	A2 ETS	A3 CRQR	A4 DR	A5 CF
Scene-understanding (language-3DGS)					
LERF [14]	✓	✗	✗	✗	△
LangSplat [28]	✓	✗	✗	✗	△
LangSplatV2 [21]	✓	✗	✗	✗	✓
Feature 3DGS [49]	✓	✗	✗	✗	✗
OpenGaussian [43]	✓	✗	✗	✗	✗
Gaussian Grouping [46]	✓	✗	✗	✗	✗
SceneSplat++ [38]	✓	✗	✗	✗	✓
VALA [42]	✓	✗	✗	✗	✗
OpenSplat3D [27]	✓	✗	✗	✗	✗
FineSplat [36]	✓	✗	✓	✗	✗
SLAG [41]	△	✗	✗	✗	✓
3DGS-based robotics					
Splat-MOVER [32]	✓	✓	✗	△	✗
LERF-TOGO [30]	✓	✓	✗	✗	✗
ManiGaussian++ [35]	✗	✓	✗	✗	△
GWM [25]	✗	✓	✗	✗	△
GaussNav [19]	✗	✓	✗	✗	✗
3DGSNav [48]	△	✓	✗	✗	✗
SAGE-3D [40]	△	✓	✗	✗	△
RoboTidy [39]	△	✓	✗	✗	✗
Vision-language-action models (no 3DGS)					
RT-2 [3]	✗	✓	✗	✗	△
OpenVLA [16]	✗	✓	✗	✗	△
3D-VLA [47]	✗	✓	✗	✗	✗
✓ totals (out of 22)	**12**	**11**	**1**	**0**	**3**

Table 1. LEGS-Bench meta-coverage matrix. ✓ = axis reported with concrete numbers on the paper's claimed scenes per the rubric in §3; △ = partial (e.g., closed-vocab where open-vocab is required, or training time without query latency); ✗ = unreported. The headline number: zero rows are complete. A4 (Dynamics Resilience) has no compliant paper at all.

Paper	Scenes	#	Dyn?
Scene-understanding			
LERF [14]	LERF-OVS	4	N
LangSplat [28]	LERF-OVS, 3D-OVS	9	N
Feature 3DGS [49]	LERF-OVS, Replica	7	N
OpenGaussian [43]	LERF-OVS, ScanNet	14	N
Gaussian Grouping [46]	LERF-OVS, custom	7	N
SceneSplat++ [38]	ScanNet/HM3D/Replica	1060	N
LangSplatV2 [21]	LERF-OVS, 3D-OVS	9	N
VALA [42]	LERF-OVS, 3D-OVS	9	N
OpenSplat3D [27]	ScanNet++, LERF-OVS	>50	N
FineSplat [36]	Fine-OVS (custom)	6	N
SLAG [41]	City-scale custom	4	N
3DGS robotics			
Splat-MOVER [32]	Custom tabletop	5	Y
LERF-TOGO [30]	Custom tabletop	8	Y
ManiGaussian++ [35]	RLBench bimanual	19t	Y
GWM [25]	RLBench+real	15t	Y
GaussNav [19]	HM3D nav	1000	N
3DGSNav [48]	MP3D/HM3D	20	N
SAGE-3D [40]	SAGE-Bench (VLN)	1000	Y
RoboTidy [39]	RoboTidy (VLA+VLN)	500	Y
VLA (no 3DGS)			
RT-2 [3]	Real tabletop	–	Y
OpenVLA [16]	Open-X	–	Y
3D-VLA [47]	RLBench+custom	–	Y

Table 2. Evaluation scenes per paper. Scene-understanding papers cluster on LERF-OVS/3D-OVS (4–9 static scenes); robot papers use RLBench/HM3D/custom (task-based). The two clusters share *zero* scenes—the structural reason no paper fills all five axes of Tab. 1 simultaneously.

4.2. Scene Coverage

Table 2 pairs each paper with its evaluation scene set, count, and static/dynamic flag. The two clusters share zero scenes—the structural reason no paper fills all five axes.

Case studies. *LangSplat vs. LERF-TOGO:* LangSplat reports 51.4% mIoU; LERF-TOGO achieves 69% grasp success—on entirely different scenes. Whether higher mIoU yields higher grasp success is unknown. *SceneSplat++* evaluates on 1060 scenes but reports A1 only; the most-scaled paper is least decision-relevant for embodied agents. *RoboTidy* reports A2 on 500 scenes but uses a fixed vocabulary, collapsing A1 to closed-set classification. Each paper made a defensible scoping decision; the gap is structural.

4.3. Adoption Cost

Making an existing paper LEGS-compliant requires ≈**10–25 H100 GPU-hours** on the ten-scene pool: A2 harness (∼1 GPU-h/scene ×10); A4 re-evaluation (3 displacement settings ×15 min/scene ≈7.5 GPU-h); A5 profiling (negligible); A3 annotation is offline. This is roughly one Mip-NeRF 360 reproduction—a budget any CVPR-scale submission already absorbs. Retraining closed-vocabulary methods is method-specific and not part of compliance cost.

5. Conclusion

No paper jointly reports open-vocabulary scene-understanding quality and downstream embodied-task success on the same 3DGS scene instances. Across twenty-two papers and 110 evaluation cells, zero rows are complete—the gap survives restriction to the three widely-measured axes. Recent benchmarks (SceneSplat-Bench [37], SAGE-Bench [40], RoboTidy [39]) each push one axis but none closes all five.

LEGS-Bench proposes five axes—scene-understanding fidelity, embodied task success, compositional query robustness, dynamics resilience, and compute footprint—on the same scene pool. Four axes run on existing infrastructure (Habitat-GS [44]); only A3 annotation requires new work.

Limitations. (a) Meta-analytic evidence inherits source-paper reporting biases. (b) Compositional-query annotations do not yet exist. (c) A4 has no compliant paper, so no calibration baseline exists. (d) Zero on A3+A4 partly measures measurement-novelty; the robust subset (§4) mitigates this. (e) Protocol utility depends on adoption; the paper ships a specification, not a leaderboard.

Call to action. We ask that future language-3DGS papers report all five LEGS-Bench axes on a shared scene set. "Which language-3DGS method is best for embodied agents?" is today ill-posed; LEGS-Bench is the minimal protocol for making it well-posed.

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